

GHAZI UNIVERSITY
D.G. KHAN

Student Name: Laiba Shareef
Roll No: 28
Department: BSCS 1st Semester
Section: C
Subject: Programming
Lab No: 02

CamScanner

Programming fundamental Lab "2"

Task 1

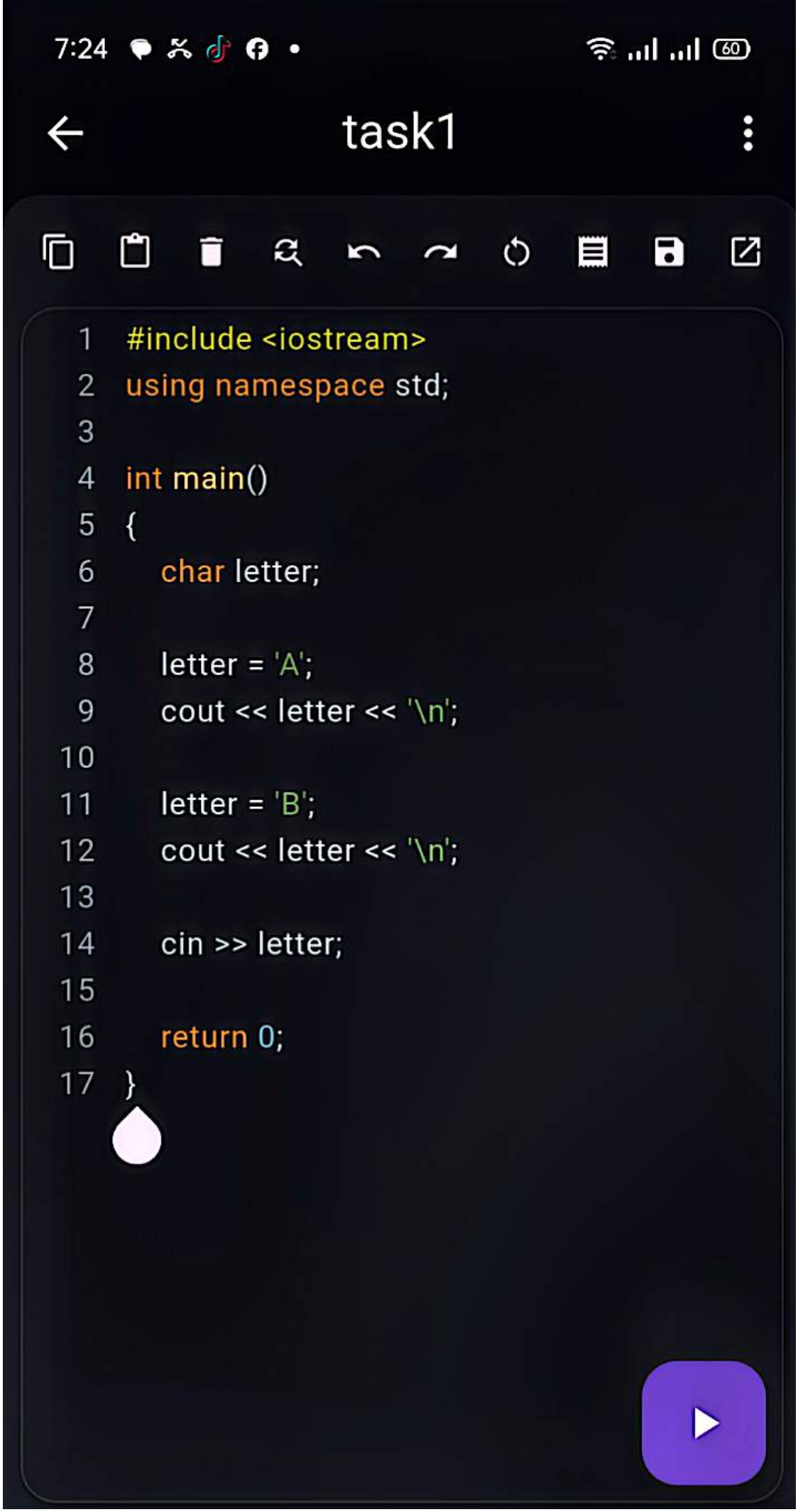
Type and save the following program

```
#include<iostream>
using namespace std;
int main( )
{
 char letter;
 letter = 'A';
cout<< letter <<'\n';
letter = 'B';
cout <<letter <<'\n';
cin >>letter;
return 0;
}
```

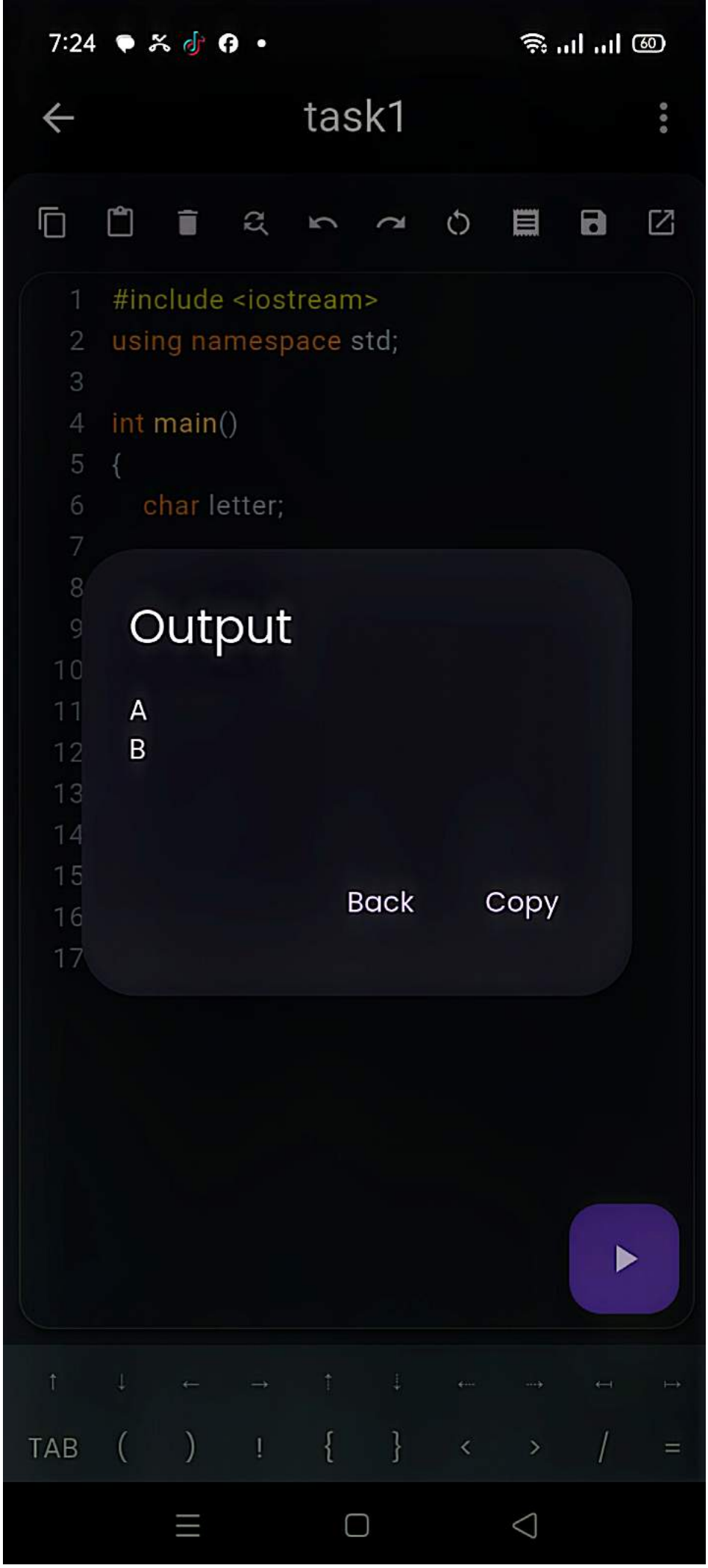
Output

A
B

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7:24
task1
#include <iostream>
using namespace std;
int main()
{
char letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}

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7:24
task1
#include <iostream>
using namespace std;
int main()
{
char letter;
Output
A
B
Back
Copy
TAB ( ) ! { } < > / =

Program 2

```
#include <iostream>
using namespace std;
int main()
{
  char letter;
  letter = 65;
  cout << letter << '\n';
  cin >> letter;
  return 0;
  }
```

output

A

Program 3

```
#include<iostream>
using namespace std;
int main()
{
  int letter;
  letter = 'A';
```

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```
cout<<letter<<'\n';
 letter = 'B';
cout<<letter<<'\n';
cin>>letter;
return 0;
}
```

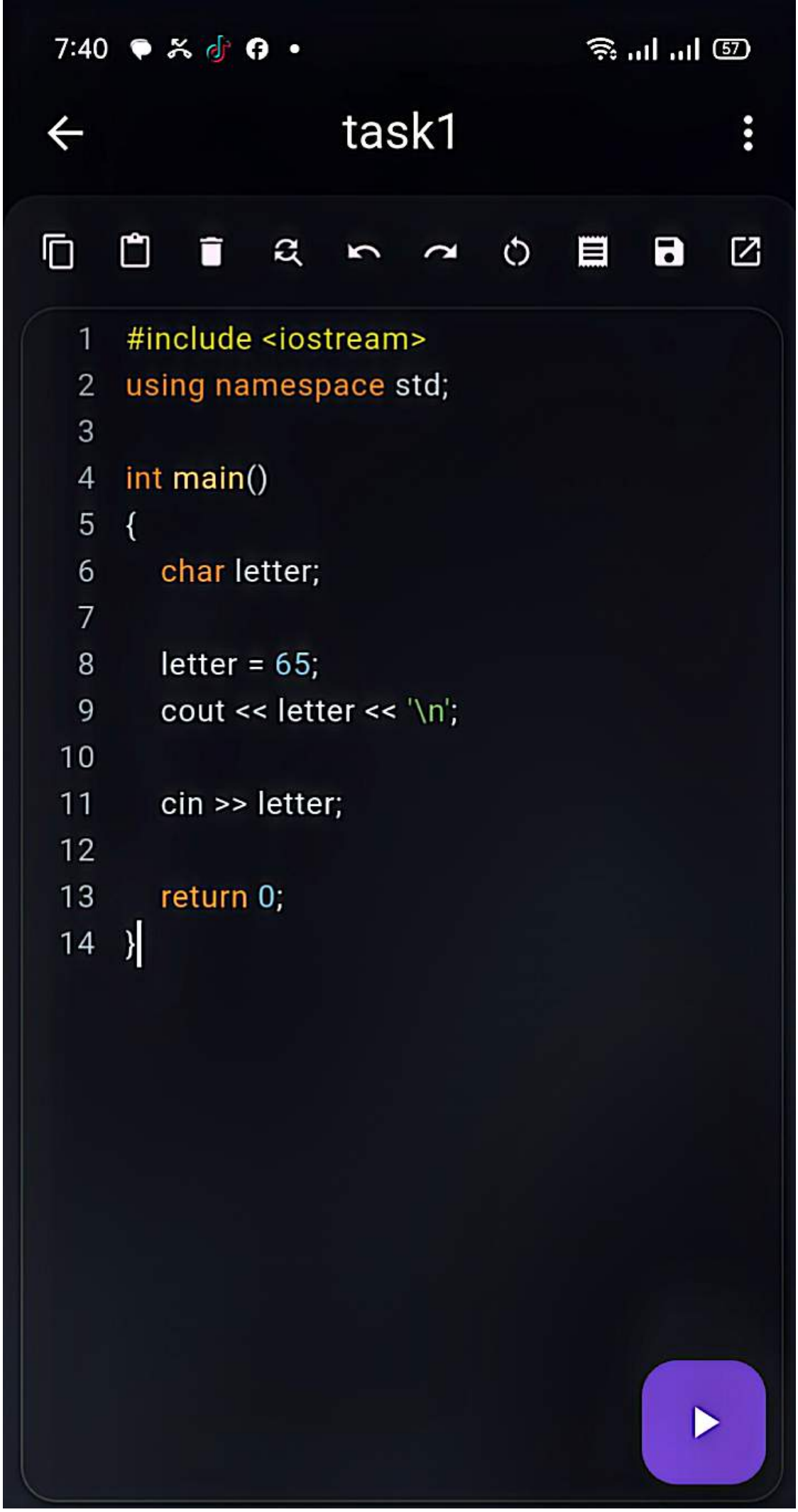
Output

65
66

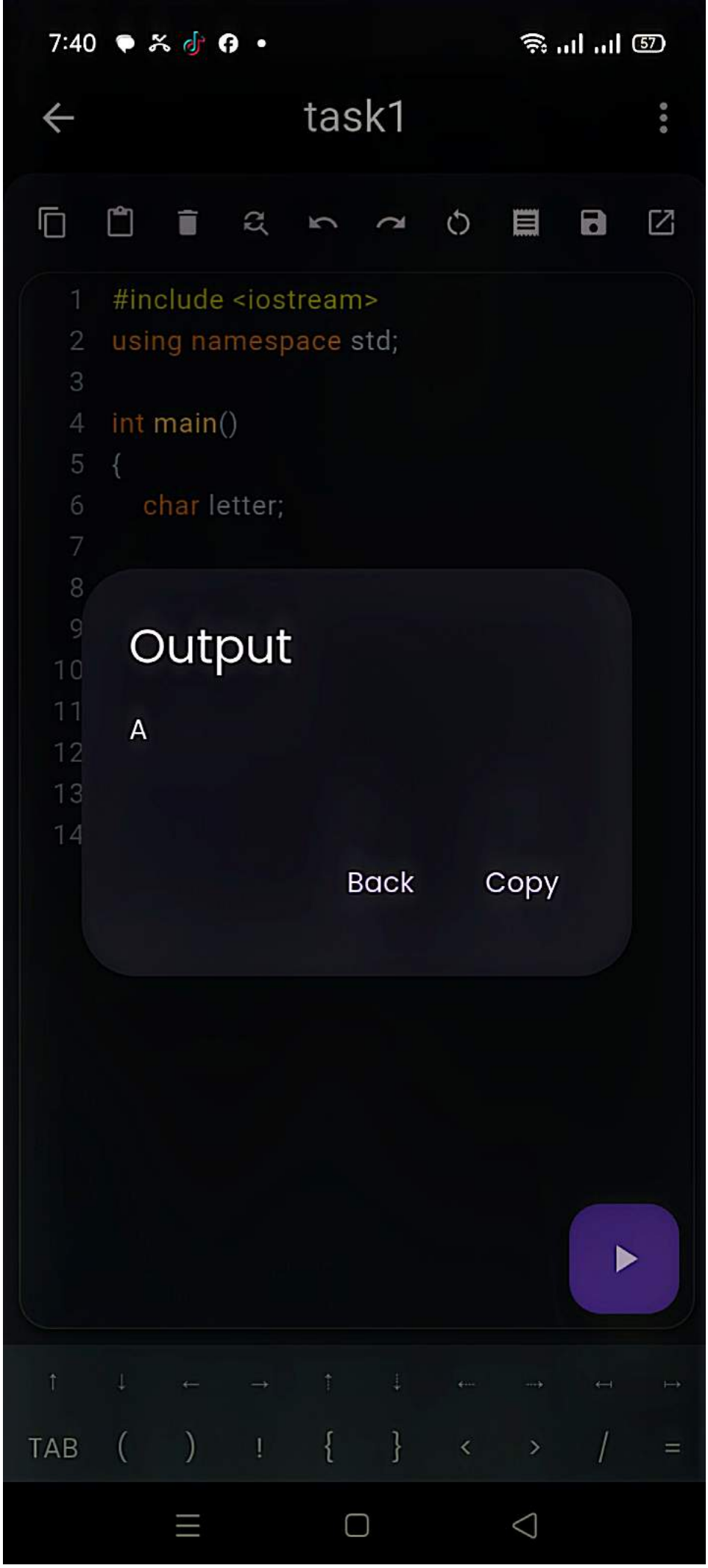
Task 2: Arithmetic Operation

```
#include<iostream>
using namespace std;
int main()
{
 int number1, num2;
cout<<"Enter two integers:";
cin >> num1 >> number2;
cout<<addition ="<<numb + num2<<endl;
cout << Subtration = " << number * num2 <<endl;
cout<< multiplication="<<num1*num2<<endl;
 cout<< division"<<num1/num2<<endl;
return 0;
}
```

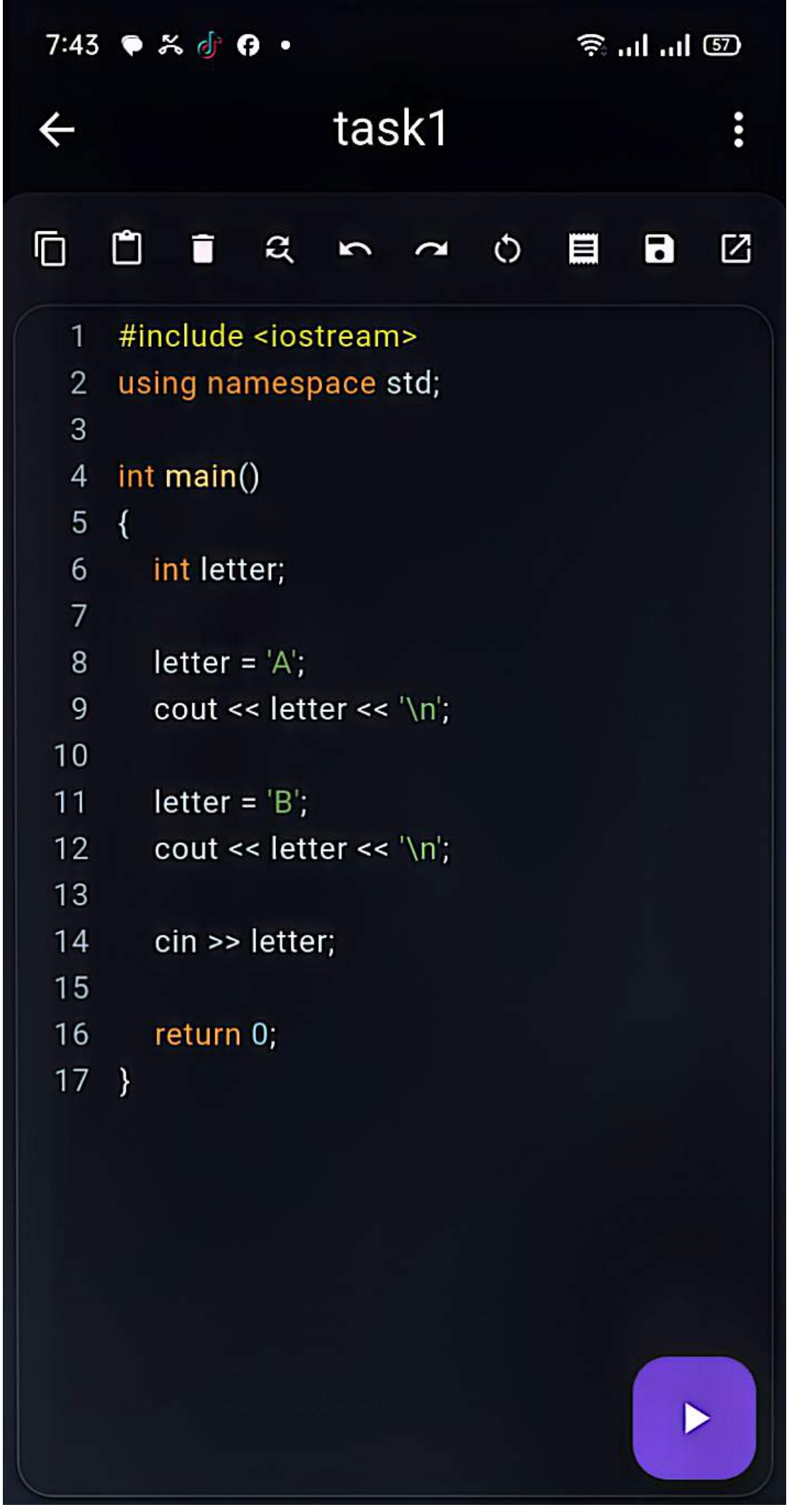
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7:40
task1
#include <iostream>
using namespace std;
int main()
{
char letter;
letter = 65;
cout << letter << '\n';
cin >> letter;
return 0;
}

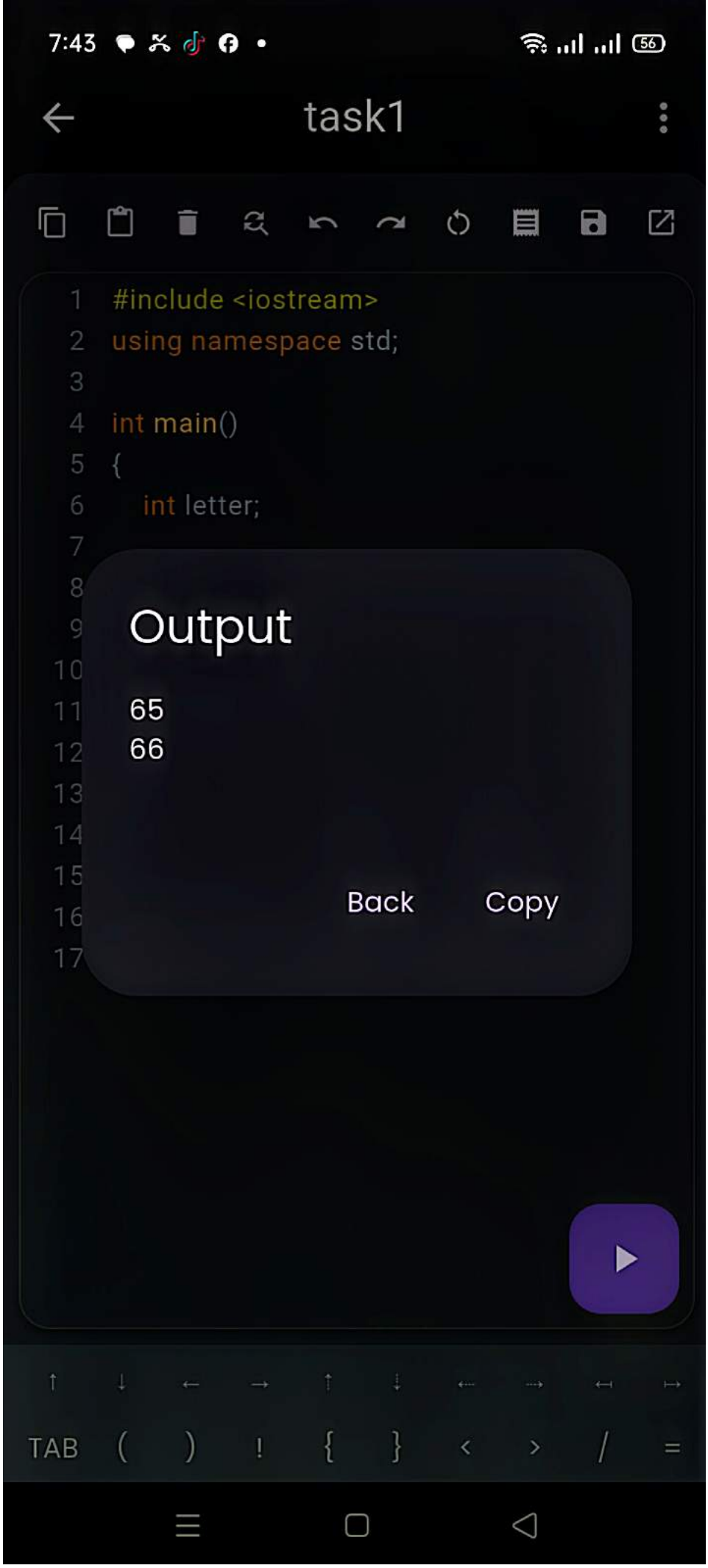
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7:40
task1
#include <iostream>
using namespace std;
int main()
{
char letter;
Output
A
Back
Copy
TAB ( ) ! { } < > / =

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7:43
task1
#include <iostream>
using namespace std;
int main()
{
int letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}

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7:43
task1
#include <iostream>
using namespace std;
int main()
{
int letter;
Output
65
66
Back
Copy
TAB ( ) ! { } < > / =

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```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;

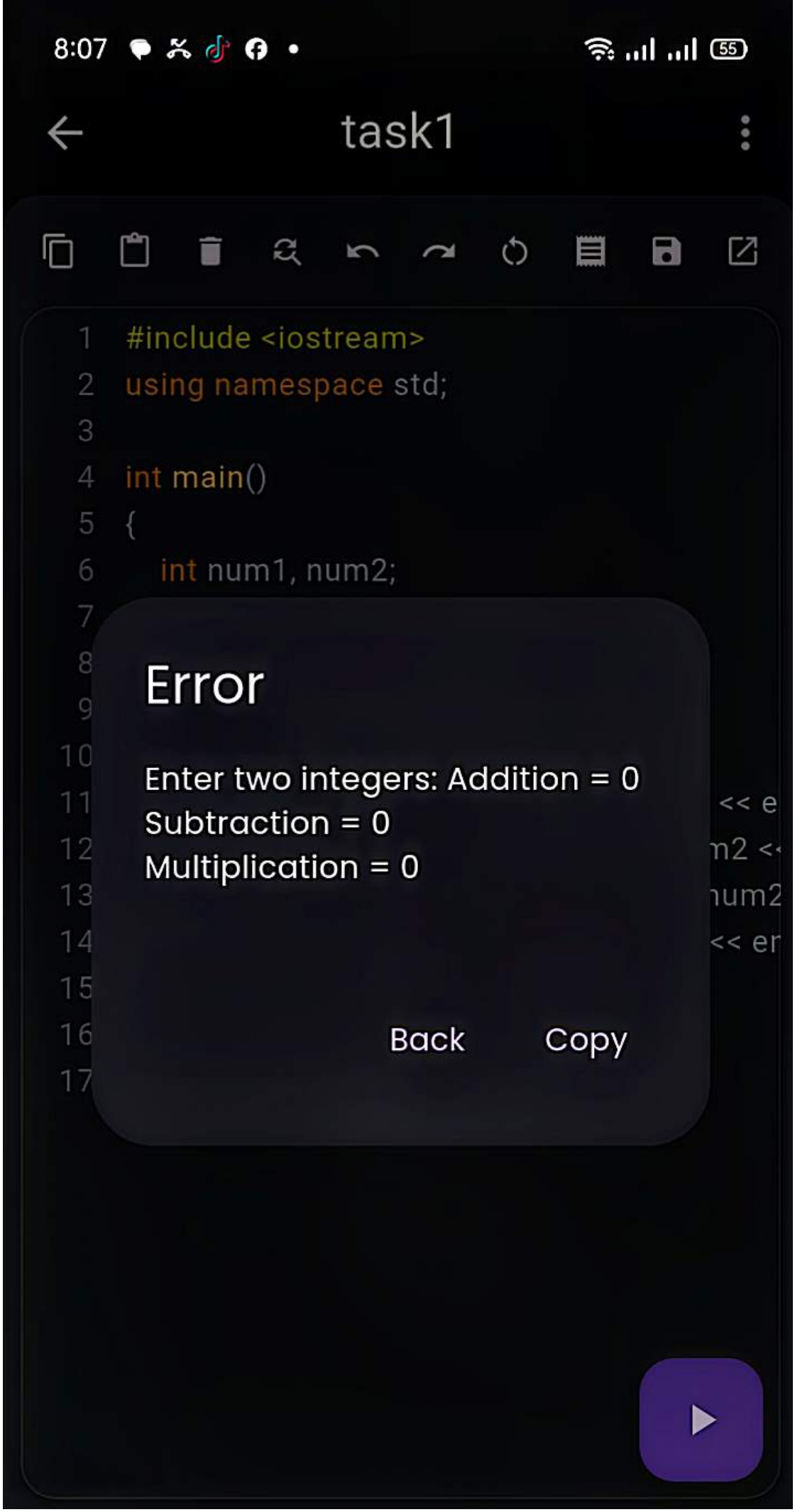
    cout << "Enter two integers: ";
    cin >> num1 >> num2;

    cout << "Addition = " << num1 + num2 << e
    cout << "Subtraction = " << num1 - num2 <<
    cout << "Multiplication = " << num1 * num2
    cout << "Division = " << num1 / num2 << er

    return 0;
}
```



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8:07
task1
#include <iostream>
using namespace std;
int main()
{
int num1, num2;
Error
Enter two integers: Addition = 0
Subtraction = 0
Multiplication = 0
Back
Copy

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Output

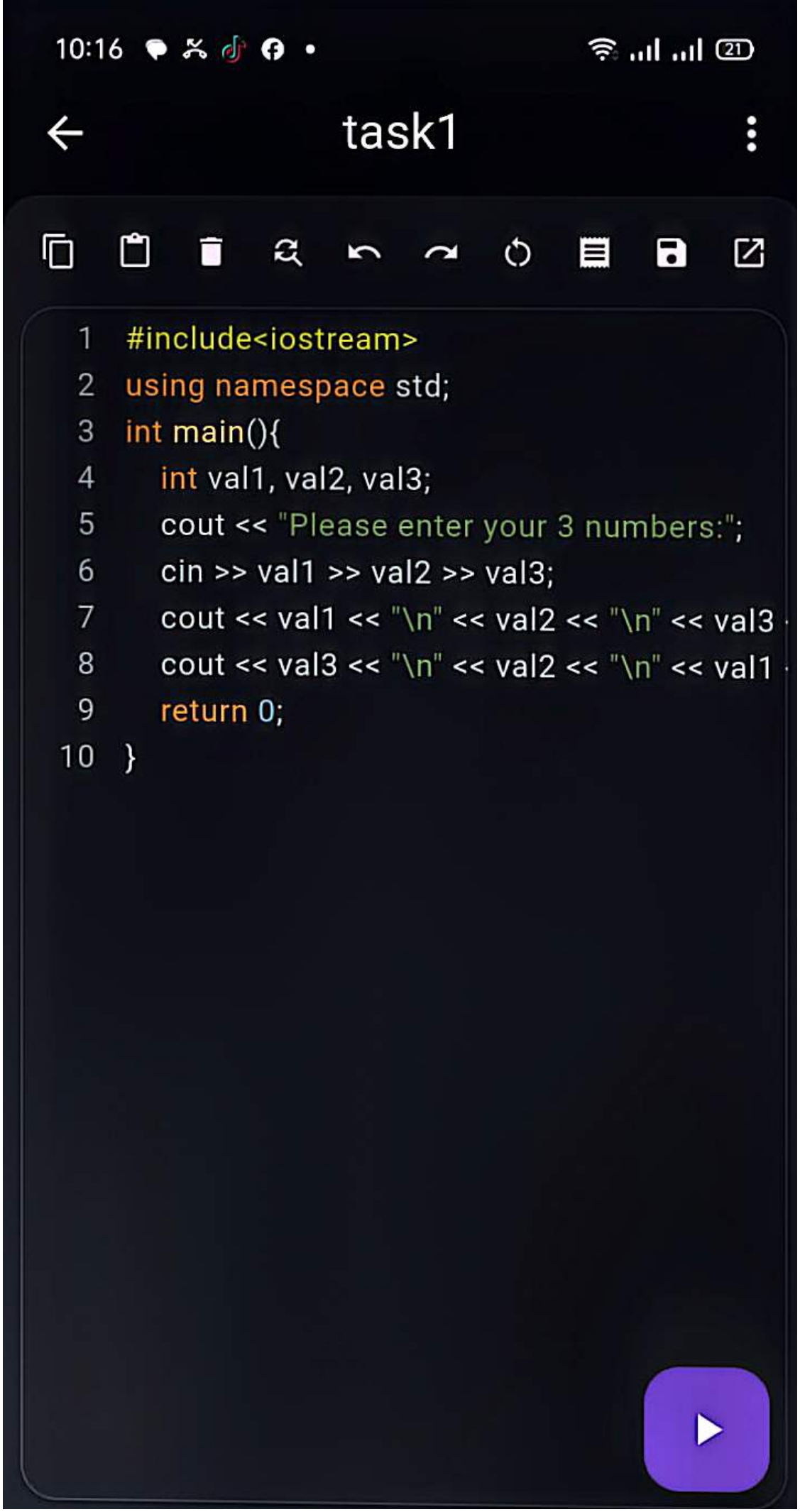
Enter two integers: Addition = ()
Subtration = ()
Multiplication = ()

Task 3 output of given code

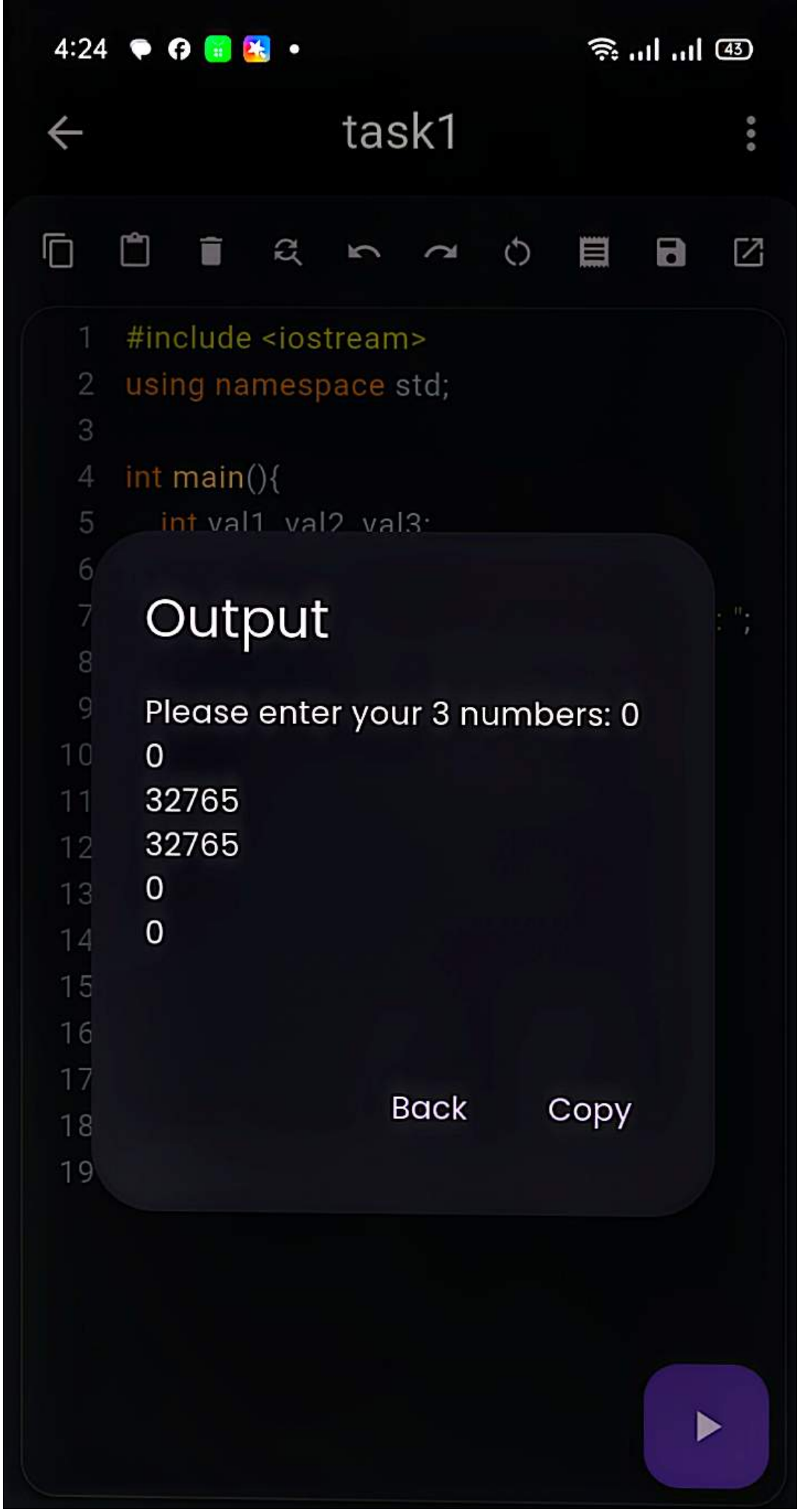
```
#include<iostream>
using namespace std;
int main( )
{
int val1;
int val2;
int val3;
cout<<" please enter your number:";
cin>>val1>>val2>>val3;
cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
Cout<<val3<<"\n<<val2<<"\n"<<val1<<"\n";
return0;
}
```

Output

10
20
30
30
20
10


10:16
task1
#include<iostream>
using namespace std;
int main(){
int val1, val2, val3;
cout << "Please enter your 3 numbers:";
cin >> val1 >> val2 >> val3;
cout << val1 << "\n" << val2 << "\n" << val3
cout << val3 << "\n" << val2 << "\n" << val1
return 0;
}

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4:24
task1
#include <iostream>
using namespace std;
int main(){
int val1, val2, val3;
Output
Please enter your 3 numbers: 0
0
32765
32765
0
0
Back
Copy

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Task 4

write a C++ program to calculate the distance between two points

formula

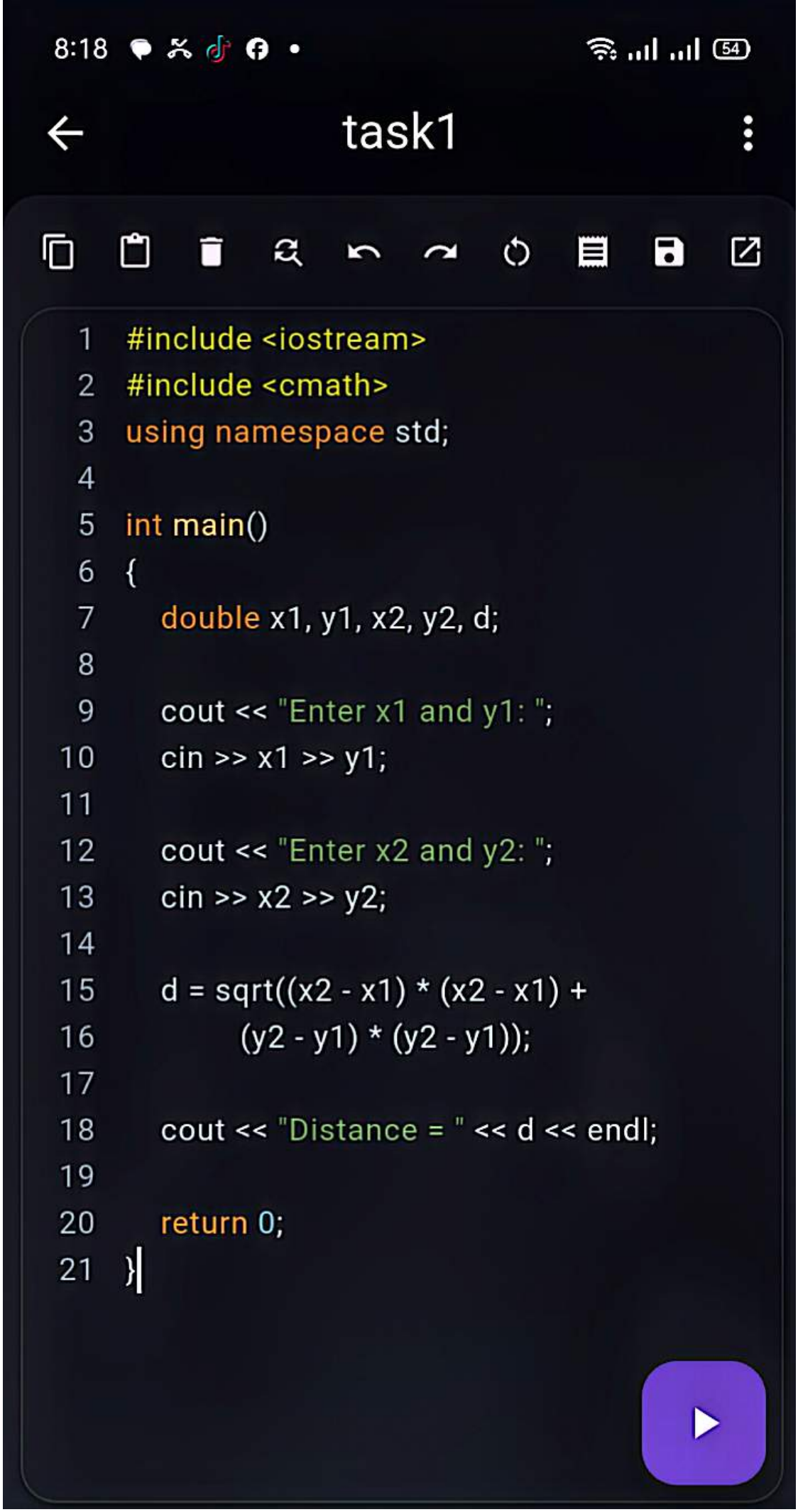
$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

```
#include<iostream>
using namespace std;
int main()
{
 double x1, y1, x2, y2;
 cout<<"enter x2 y3 x5 y7:";
 cin>>x1>>y1>>x2>>y2;
 double d=pow(x2-x1,2)+pow(y2-y1,2
 cout<<"distance="<<d;
 return 0;
}
```

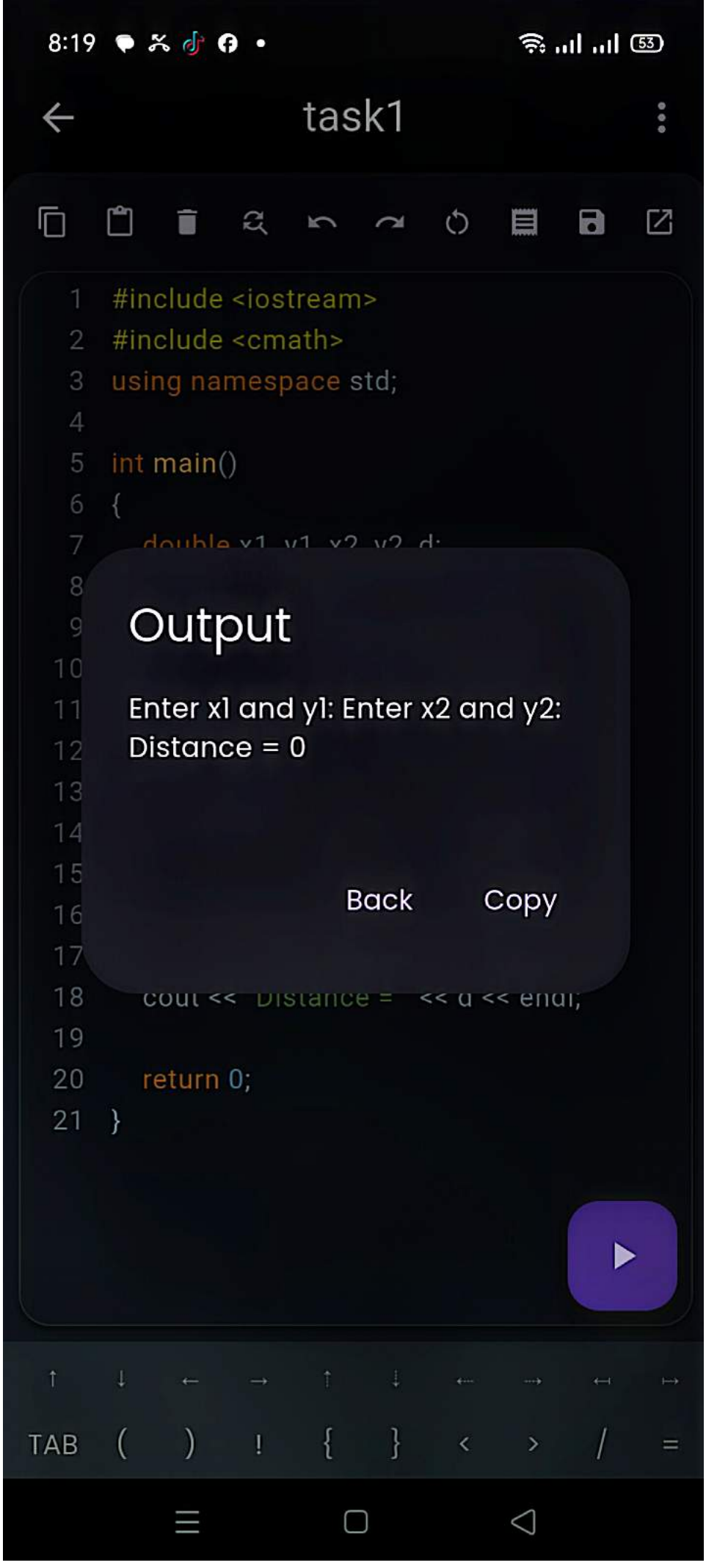
output

Enter x1 and y1 : Enter x2 and y2

distance = 0


8:18
54
task1
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
double x1, y1, x2, y2, d;
cout << "Enter x1 and y1: ";
cin >> x1 >> y1;
cout << "Enter x2 and y2: ";
cin >> x2 >> y2;
d = sqrt((x2 - x1) * (x2 - x1) +
(y2 - y1) * (y2 - y1));
cout << "Distance = " << d << endl;
return 0;
}

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8:19
task1
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
Output
Enter x1 and y1: Enter x2 and y2:
Distance = 0
Back
Copy
cout << "Distance = " << d << endl;
return 0;
}
TAB ( ) ! { } < > / =

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```
>> length >> width;
area = length * width;
cout << area;
return 0;
}
```

Output

Enter length and width: 10 5.50

Task 5 Area of a triangle

```
#include <iostream>
using namespace std;
int main()
{
  float base, height, area;
  cin >> base >> height;
  area = base * height * 0.5;
  cout << area;
return 0;
}
```

Output

Enter base and height: 10 6 3

8:26

task1

```cpp
#include <iostream>
using namespace std;

int main()
{
    double r, area;

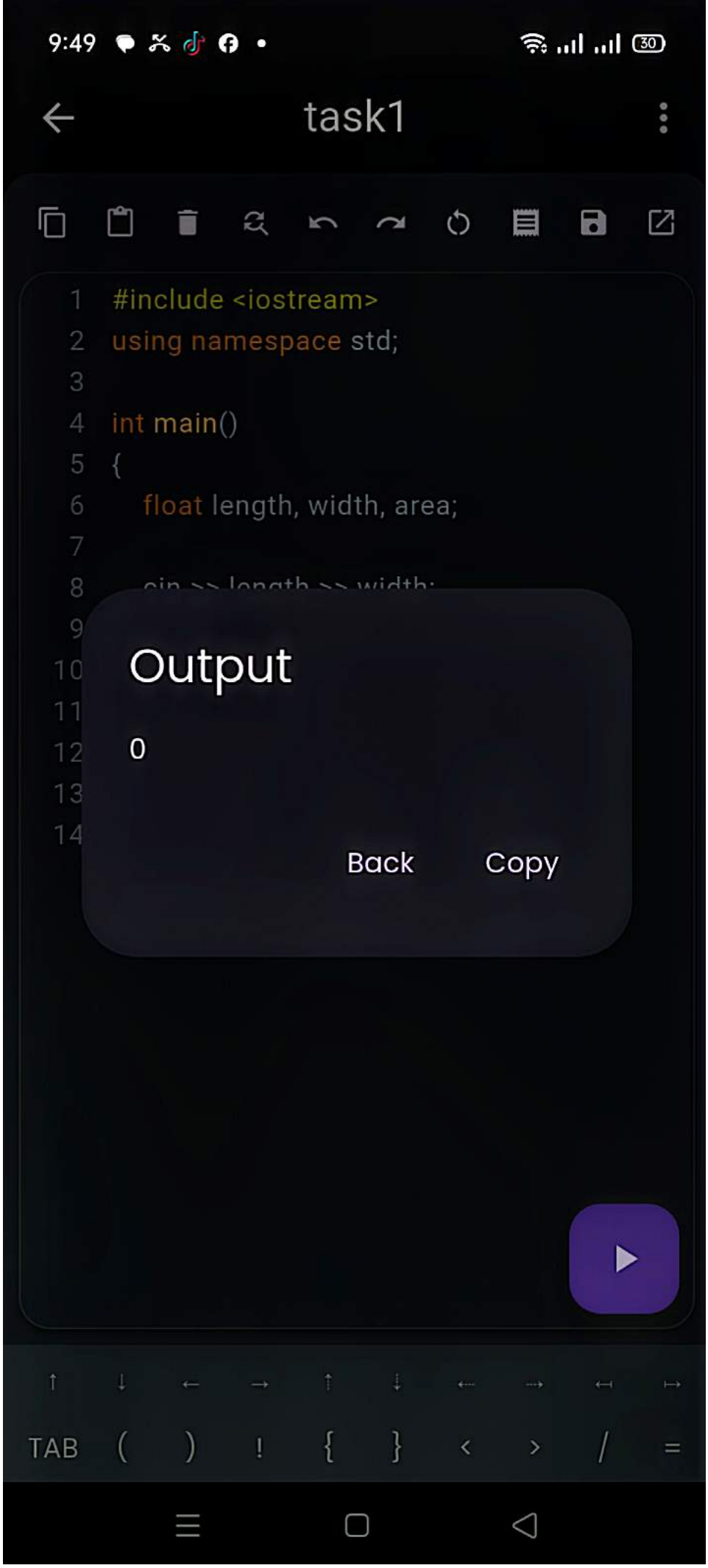
    cout << "Enter radius: ";
    cin >> r;

    area = 3.14159 * r * r;

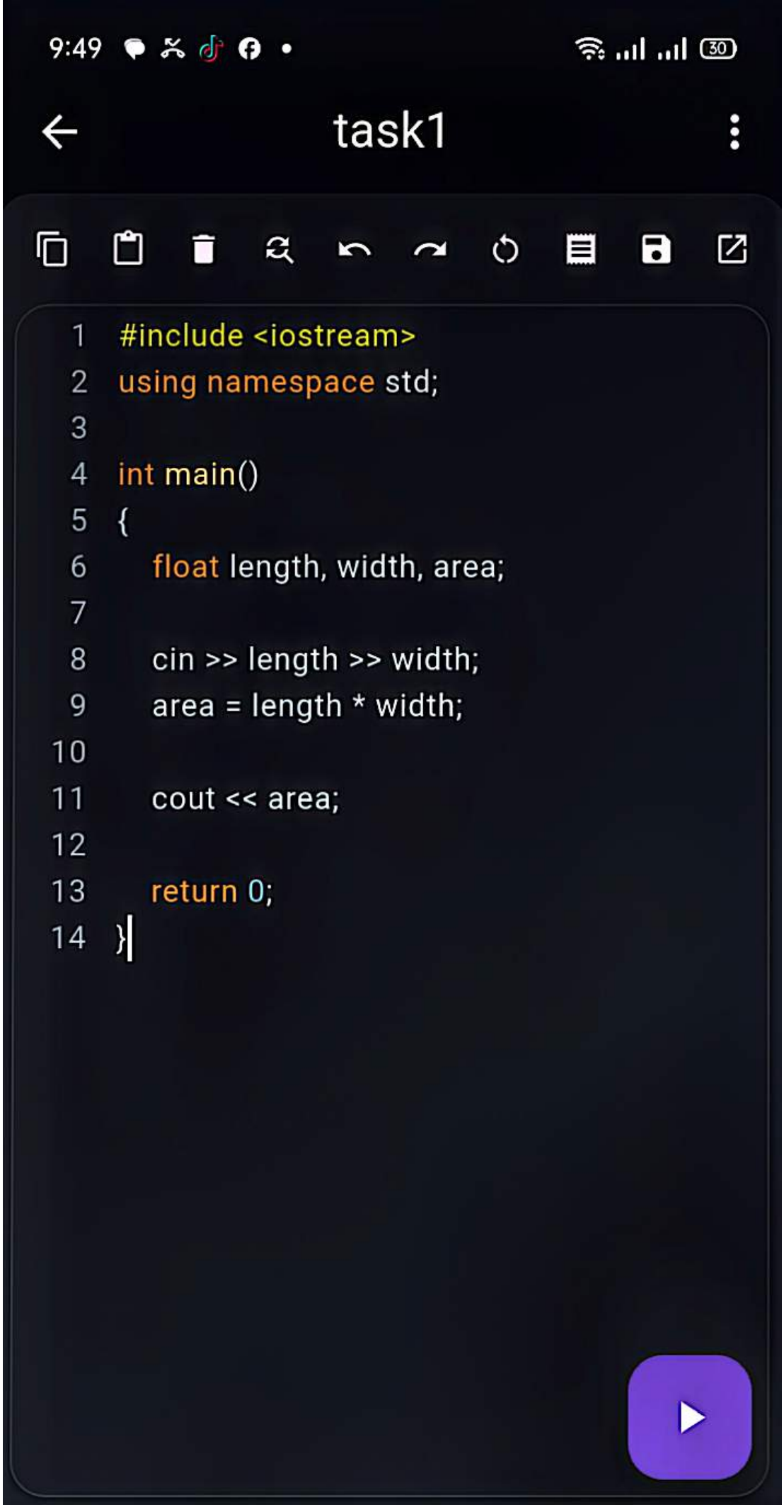
    cout << "Area of Circle = " << area << endl;

    return 0;
}
```

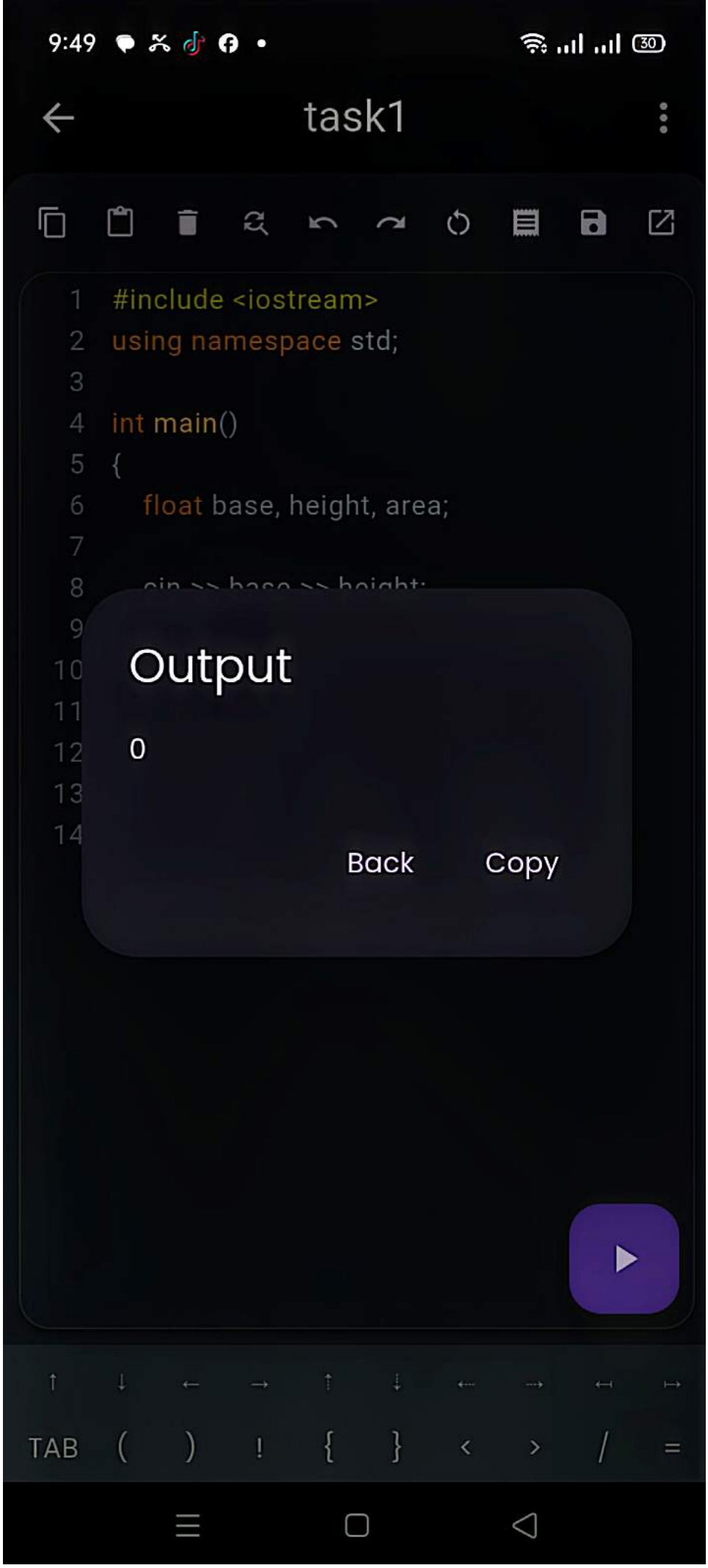
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9:49
task1
#include <iostream>
using namespace std;
int main()
{
float length, width, area;
Output
0
Back
Copy
TAB ( ) ! { } < > / =

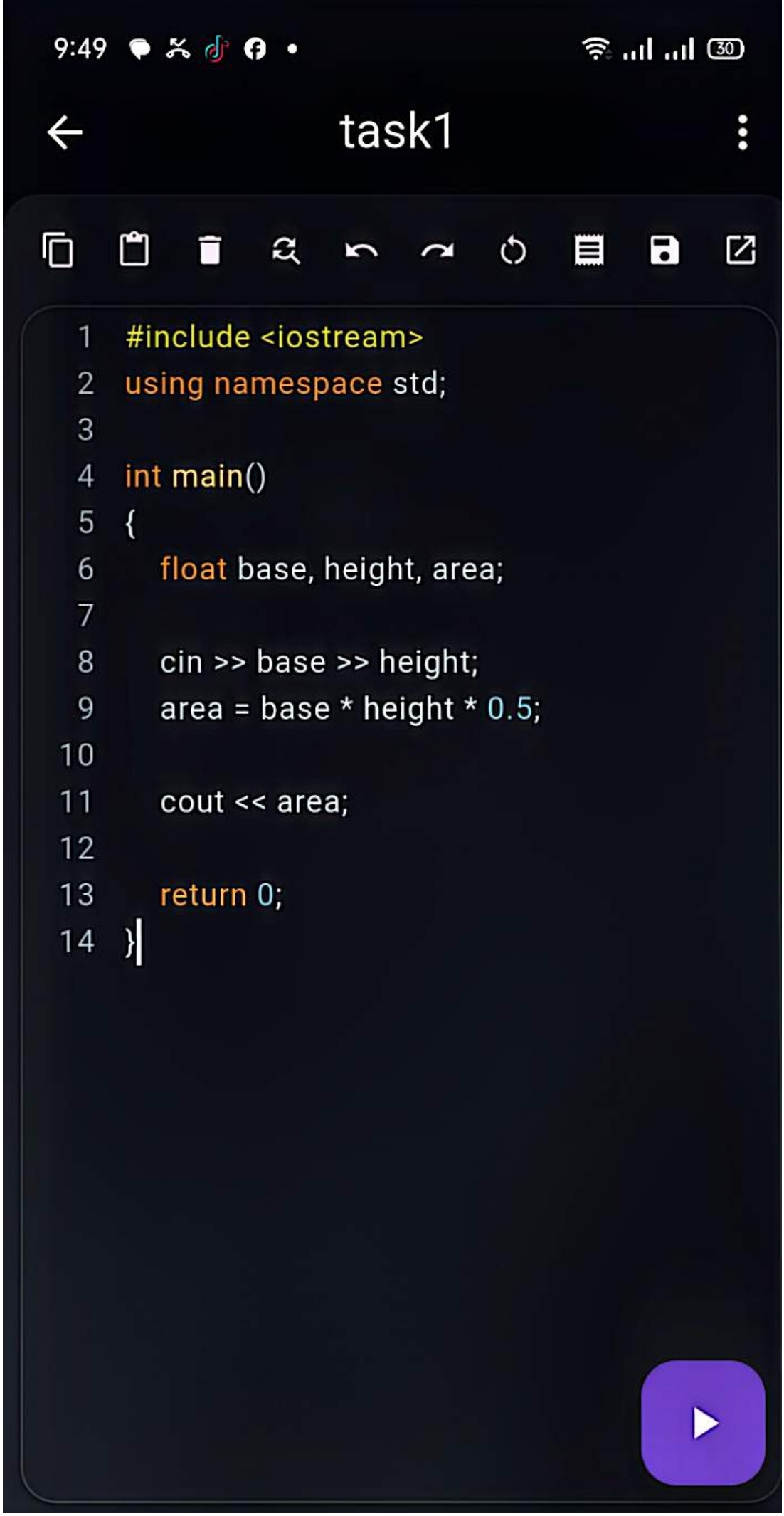
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9:49
task1
#include <iostream>
using namespace std;
int main()
{
float length, width, area;
cin >> length >> width;
area = length * width;
cout << area;
return 0;
}

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9:49
task1
#include <iostream>
using namespace std;
int main()
{
float base, height, area;
Output
0
Back
Copy
TAB ( ) ! { } < > / =

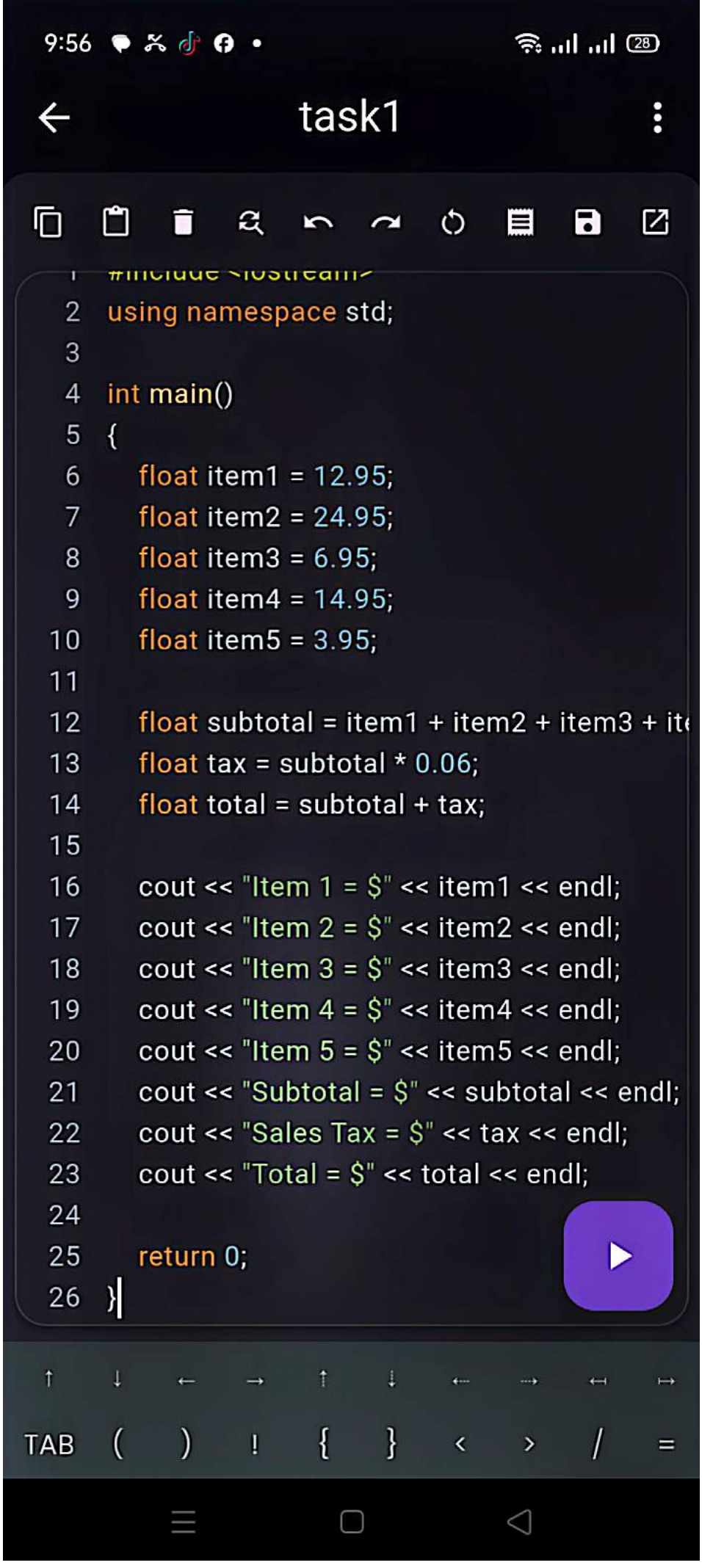
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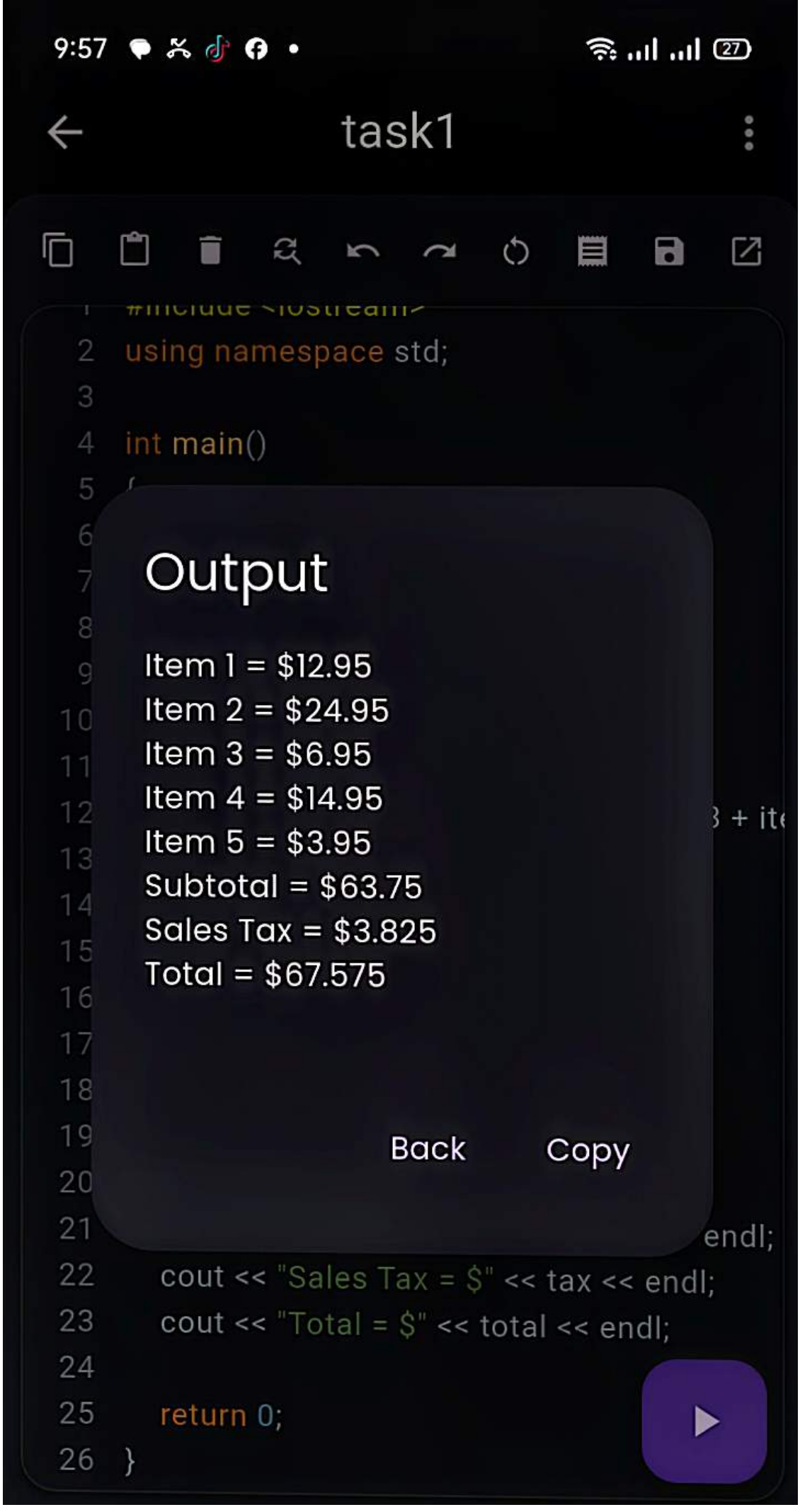

9:49
task1
#include <iostream>
using namespace std;
int main()
{
float base, height, area;
cin >> base >> height;
area = base * height * 0.5;
cout << area;
return 0;
}

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...sk 6—Total purchase

```
#include <iostream>
using namespace std;
int main()
{
  float item1 = 12.95;
  float item2 = 24.95;
  float item3 = 6.95;
  float item4 = 14.95;
  float item5 = 3.95;
float subtotal = item1+item2+item3
 float tax = subtotal * 0.06;
float total = subtotal + tax;
cout << "Item1 = $" << item1 << endl;
cout << "Item2 = $" << item2 << endl;
 cout << "Item 3 = $" << item3 << endl;
 cout << "Item 4 = $" << item4 << endl;
 cout << "Item 5 = $" << item5 << endl;
 cout << subtotal = $" << subtotal << endl;
 cout << "sales tax = $" << tax << endl;
 cout << "total = $ " << total << endl;
return 0;
```


9:56
task1
#include <iostream>
using namespace std;
int main()
{
float item1 = 12.95;
float item2 = 24.95;
float item3 = 6.95;
float item4 = 14.95;
float item5 = 3.95;
float subtotal = item1 + item2 + item3 + it
float tax = subtotal * 0.06;
float total = subtotal + tax;
cout << "Item 1 = $" << item1 << endl;
cout << "Item 2 = $" << item2 << endl;
cout << "Item 3 = $" << item3 << endl;
cout << "Item 4 = $" << item4 << endl;
cout << "Item 5 = $" << item5 << endl;
cout << "Subtotal = $" << subtotal << endl;
cout << "Sales Tax = $" << tax << endl;
cout << "Total = $" << total << endl;
return 0;
}
TAB ( ) ! { } < > / =


9:57
task1
using namespace std;
int main()
Output
Item 1 = $12.95
Item 2 = $24.95
Item 3 = $6.95
Item 4 = $14.95
Item 5 = $3.95
Subtotal = $63.75
Sales Tax = $3.825
Total = $67.575
Back
Copy
endl;
cout << "Sales Tax = $" << tax << endl;
cout << "Total = $" << total << endl;
return 0;
}

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Input

Item1 = $12.95
Item2 $24.95
Item3 = $695
Item4: $1495
Item5 = $3.95
Subtotal = $63.75
Sales tars $3.825
Total s $67.575

Bonus task – break Number

```
#include <iostream>
using namespace std;
int main( )
{
 int num;
 cin >> num;
 cout << num /10000<<" 1";
 cout <<(num/10000%10<<" 2 ";
 cout <<(num/100 %.10<<" 3 ";
 cout <<(num /10) %.10<<"4 ";
 cout <<num%.10;
 return 0;
}
```

```cpp
#include <iostream>
using namespace std;

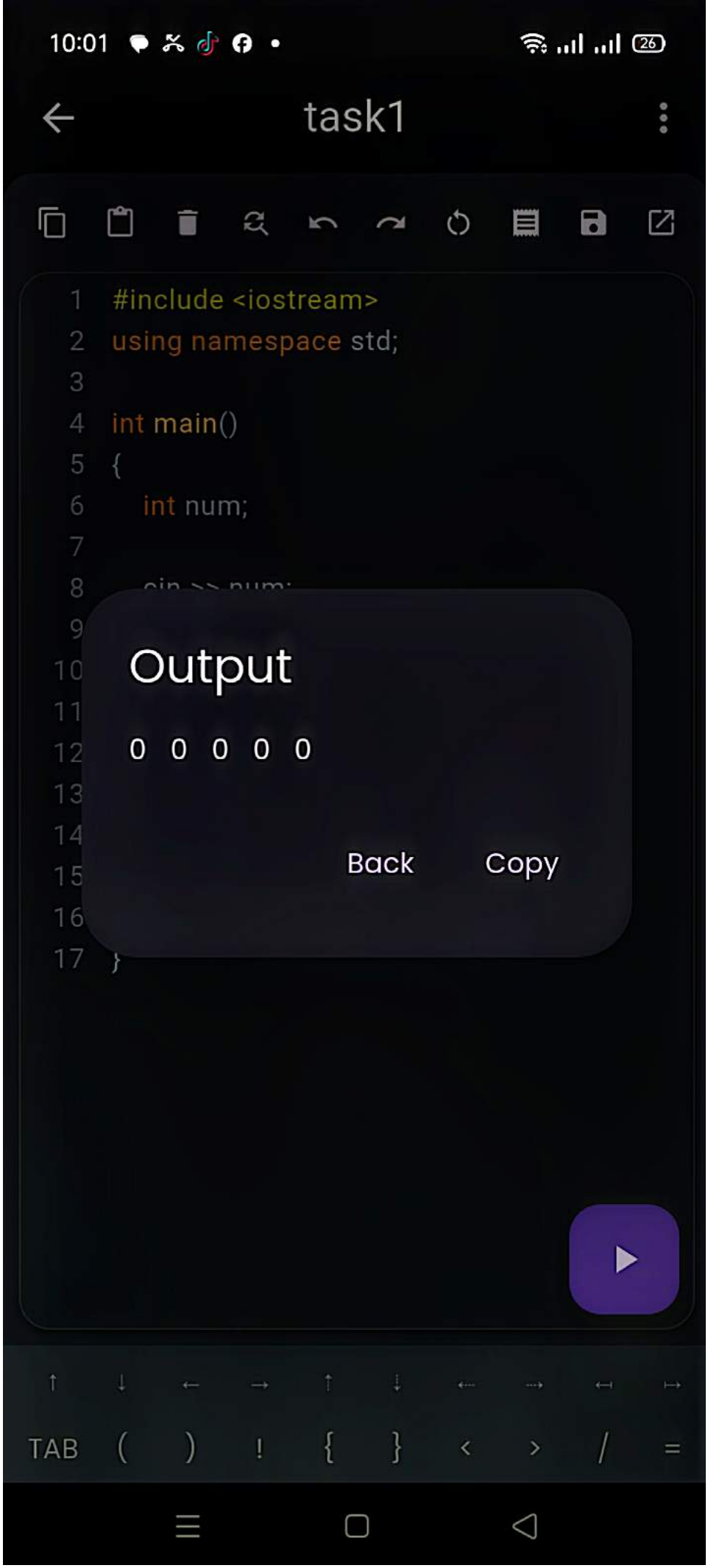
int main()
{
    int num;

    cin >> num;

    cout << num / 10000 << "   ";
    cout << (num / 1000) % 10 << "   ";
    cout << (num / 100) % 10 << "   ";
    cout << (num / 10) % 10 << "   ";
    cout << num % 10;

    return 0;
}
```

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10:01
task1
#include <iostream>
using namespace std;
int main()
{
int num;
Output
0 0 0 0 0
Back
Copy
}
TAB ( ) ! { } < > / =

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Input

1 2 3 4

Home task 1 — Exchange:

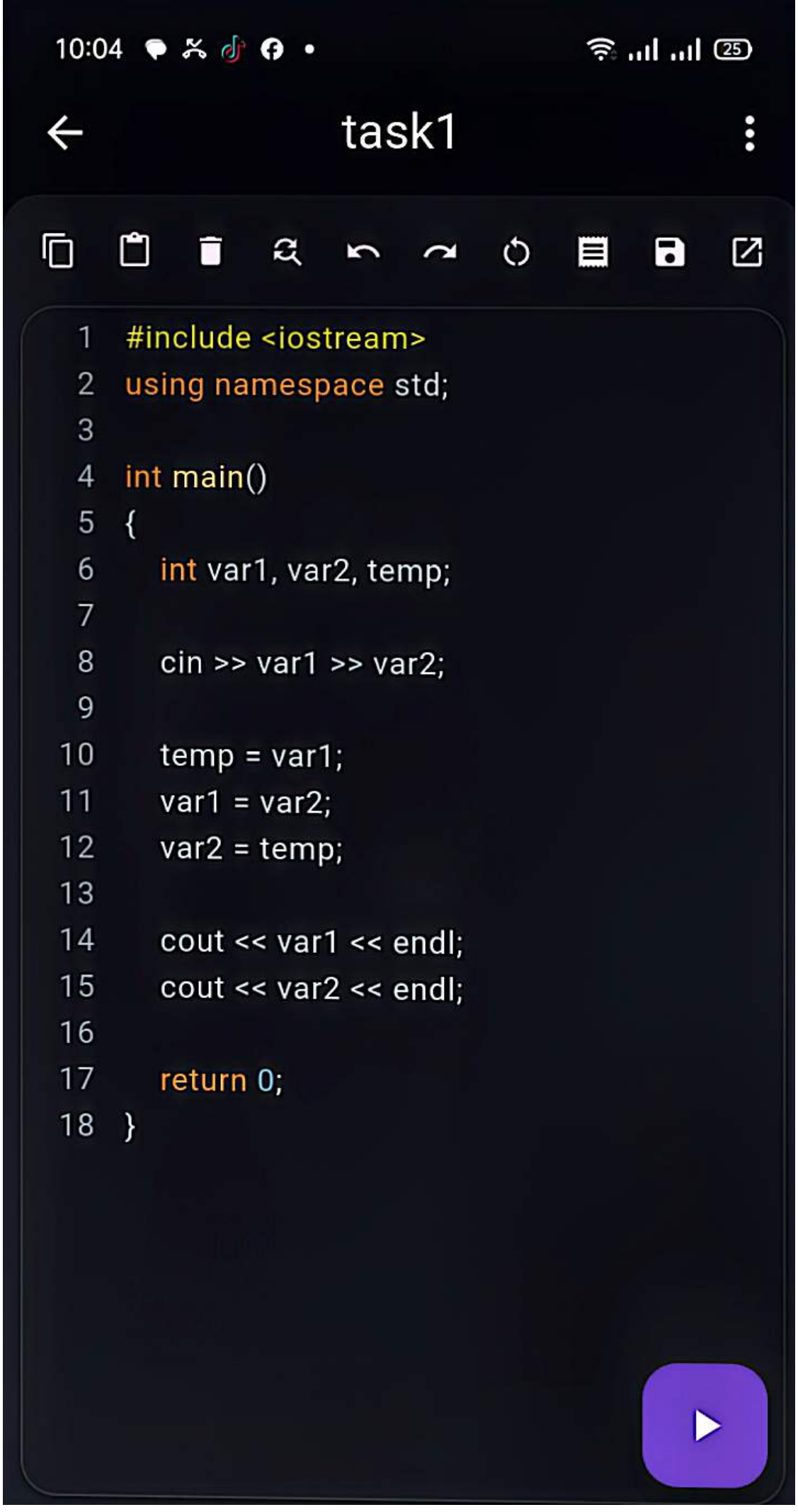
```
include <iostream>
Using namespace std;
int main( )
{
  int val, val2 temp;
cin >> val >> val2;
temp= val;
val1 = val2;
val2 = temp;
cout << val << endl;
cout<<val2<<endl;
return 0;
}
```

Output

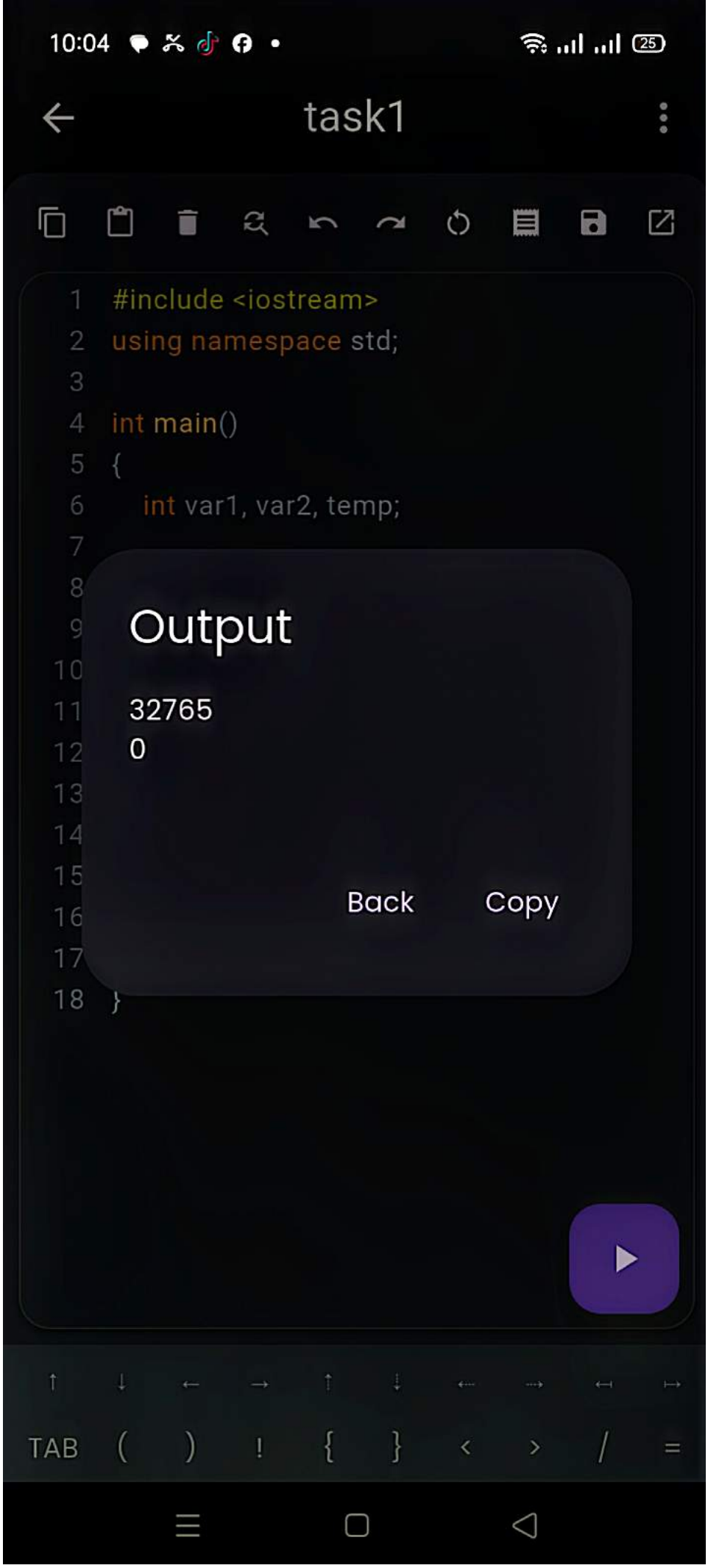
10 20

20

10


10:04
task1
#include <iostream>
using namespace std;
int main()
{
int var1, var2, temp;
cin >> var1 >> var2;
temp = var1;
var1 = var2;
var2 = temp;
cout << var1 << endl;
cout << var2 << endl;
return 0;
}

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10:04
task1
#include <iostream>
using namespace std;
int main()
{
int var1, var2, temp;
Output
32765
0
Back
Copy
}
TAB ( ) ! { } < > / =

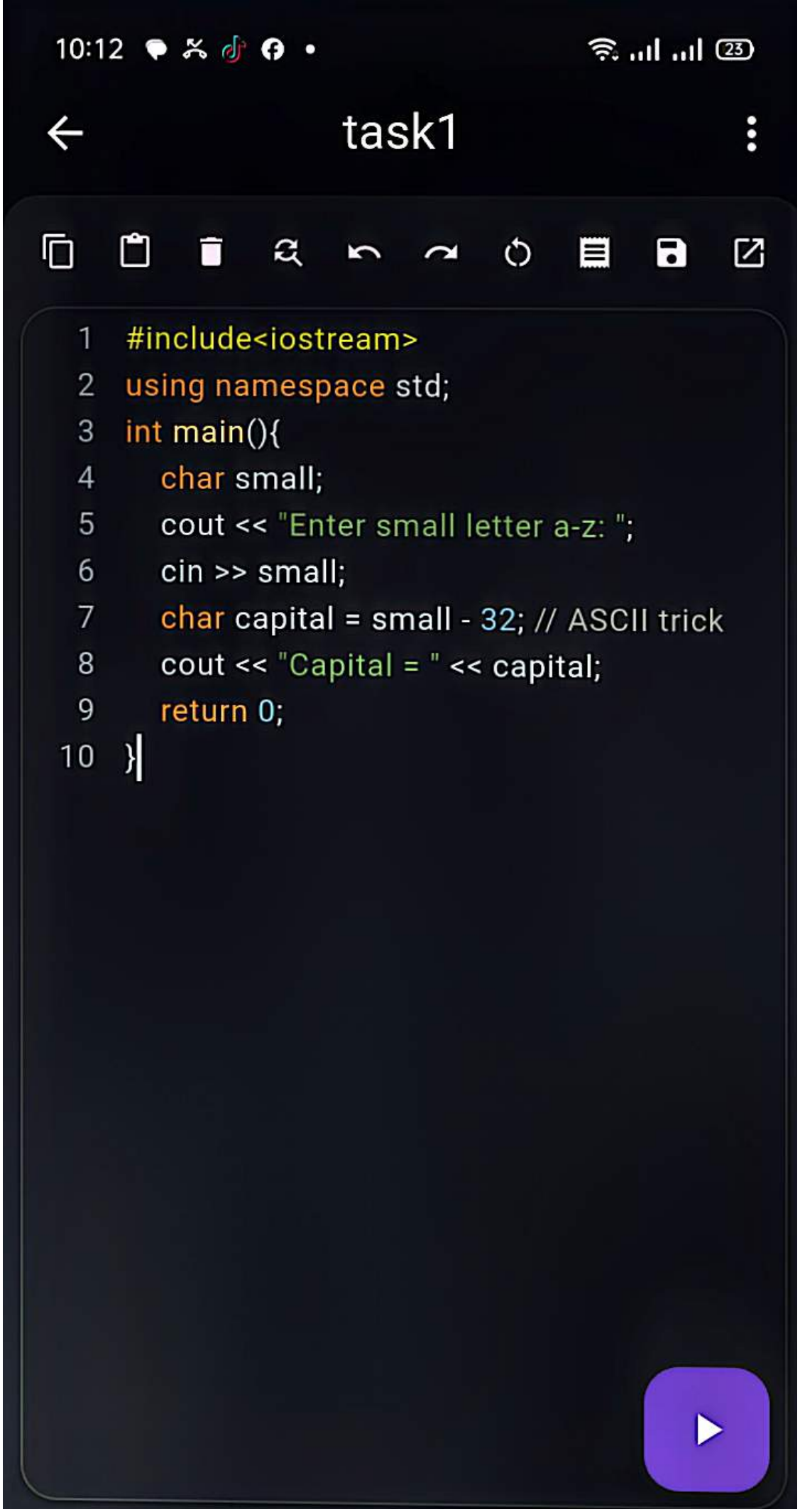
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Home task Small to capital

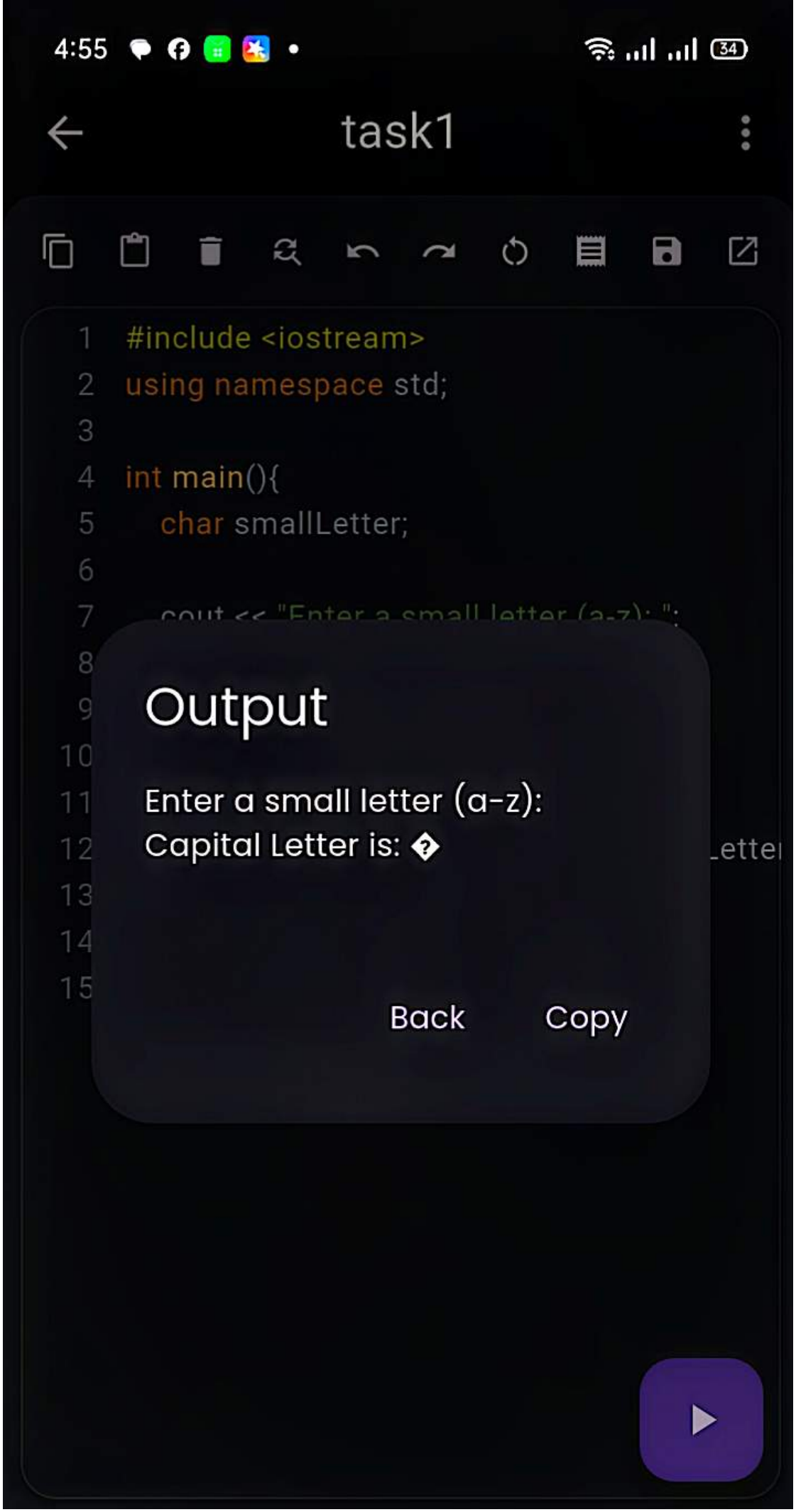
```
#include<iostream>
using namespace std;
int main()
{
 char small;
cout<<"Enter small letter a-z:";
 Cin>>small;
 char capital=small-32; // ASCII tri
 cout<<capital="<<capital;
 return 0;
 }
```

Output

```
capital=A
capital A
Enter small letter a-z:2
  capital=Z
```


10:12
task1
#include<iostream>
using namespace std;
int main(){
char small;
cout << "Enter small letter a-z: ";
cin >> small;
char capital = small - 32; // ASCII trick
cout << "Capital = " << capital;
return 0;
}

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4:55
task1
1 #include <iostream>
2 using namespace std;
3
4 int main(){
5 char smallLetter;
6
7 cout << "Enter a small letter (a-z): ";
8
9
10
11
12
13
14
15
Output
Enter a small letter (a-z):
Capital Letter is: ?
Back
Copy

CamScanner